AS/400 Modernization Blueprint

System: GAEB — General Authority for Educational Buildings (Egypt), national AS/400 (IBM i V6R1) platform, library NEWLIBGAEB
Prepared by: Veebase — Proof of Concept, November 2026 | **Status:** Working POC delivered with this blueprint

1. Executive summary

The delivered archive is not a data extract — it is the **complete source code** of GAEB's core operational system: 502 unique source members (SEU listings) covering six subsystems, plus screen specifications, ERD diagrams and reference-data exports. The system is still live: Real Estate module screenshots in the delivery are dated August 2025.

We parsed every listing programmatically and reconstructed the system's data model and process logic. From that single source of truth we have already generated the modern database schema (**125 PostgreSQL tables, 1,107 mapped fields**) and built a working bilingual web application covering all six modules plus a Migration Center. This blueprint is the ideal path to full delivery: it is a **rebuild on a modern stack with automated schema conversion and staged data migration** — not a screen-scraping or emulation exercise, and not a risky big-bang cutover.

502 source members analyzed | **169** data files (135 PF + 34 LF)
1,134 legacy fields parsed | **333** programs (270 RPG)
125 modern tables generated | **6** subsystems

2. What the legacy system actually is

Subsystem	Files / Programs	What it does
School Map (الخريطة المدرسية)	47 files / 84 pgms	National registry of every educational building: spaces, gates, fences, utilities, coordinates (GIS), areas, annexes, hosted/merged/closed schools — plus the planning engine (future-plan targets, school needs, Grade-6 solutions, GIS distance inquiry).
Budget (الموازنة)	43 files / 110 pgms	Appropriations, allowances and payments by fiscal year, governorate, work code and funding source; statement reporting. Core entry is one cloned program family (CBA009OR and ~8 variants).

Purchases & Contracts (المشتريات)	16 files / 16 pgms	The Egyptian tender lifecycle as separate controlled steps: booklet → bids → exclusion → envelope committee → financial evaluation → award committee → negotiation → award, over the shared supplier master PBYSUPL.
Technical Abstracts (المستخلصات الفنية)	25 files / 25 pgms	Contractor payment certificates (construction + maintenance). Line value = measured quantity × contract price; typed deductions (CUTKND: retention, advances, penalties); certification → billing (BIL001–BIL010) → payment (PAG*) are separate steps.
Estimates / BOQ (المقايسات)	— / 38 pgms	Priced estimates built only from the standard item catalogue (DES* masters via CAT005RR); draft (ADM*) and reviewed (ADMR*) estimates are physically separate, each with its own print program.
Real Estate (الملكية العقارية)	38 files / 60 pgms	Rented buildings and their acquisition pipeline (study → committee → purchase / negotiation / donation / continued rent / return / expropriation); expropriation cases gated by the ministerial decree (REMV01 pre-decree → REMV02 post-decree); land parcels with borders, utilities, documents and committee approvals.

One shared foundation: the building register (`PBLBULN0/PBLSCH0L`), code tables (`PBLTAB01–09`, `PBLCNTRF`, `PBLBRAL1`) and supplier master (`PBYSUPL`) are used by all six subsystems. **The money flow crosses subsystems:** Estimates → Abstracts certification → Budget payment. Geography is fully coded: 27 governorates, 7 regional centres, ~388 centres, 6,116 villages, ~35,449 village dependencies, ~292 educational administrations, 45 education stages.

3. Target architecture

- **Platform:** modern web application (the delivered POC stack), bilingual Arabic/English with full RTL, role-based access (Admin, Budget, Contracts, Technical, Viewer), complete audit log — replacing 5250 green screens and hand-built CL/RPG menu dispatchers (CAINGHIR, CBA707IR) with real navigation and permissions.
- **Database:** PostgreSQL, one schema per subsystem (`school_map`, `budget`, `purchases`, `abstracts`, `estimates`, `real_estate`, `reference`), generated from the DDS definitions (Section 4). Legacy composite keys (regional centre + branch + serial; BLCD building code) are preserved as natural keys so every migrated record stays traceable to its AS/400 origin.
- **Collapse the clones:** each cloned program family (Budget's CBA009OR family, the INV001AR1/INV005AR pair, the PMPT200R4/PMP55555 print pair) becomes **one parameterized module** in the new system. This is where the maintenance burden dies.
- **Business rules as explicit services:** the rules that today live inside RPG — building-number check digit (CHKDIG), quantity × price valuation, typed deductions, the 5-day deadline check (CHK5DAYS), decree-gated expropriation workflow, catalogue-only estimate building, draft-vs-reviewed estimate separation, tender-stage sequencing — are re-implemented as named, tested rule services, not buried in screen code.
- **GIS:** building coordinates (SCHCOORD X/Y/Z) move to PostGIS; the "schools within a distance" inquiry and the registered-vs-survey area comparison (±10% tolerance, COMGPS1R) become map features.

- **Documents & reports:** legacy printer files (the CBA333*/CBA334* statements, BOQ prints, approval prints like GRN002IP) become PDF report templates generated from the same data.
- **Deployment:** cloud, private hosting on Veebase infrastructure, or fully on-premise/air-gapped at the Authority — the same options Veebase offers across its platform.

4. Data migration — the automated path (already proven in the POC)

Step 1 — Schema conversion (done)

Every DDS physical-file definition was parsed from the listings (record formats, field names, lengths, decimals, keys, UNIQUE) into a machine inventory (`legacy_inventory.json`), then converted to PostgreSQL DDL: **125 tables, 1,107 fields**, with a field-mapping register (`field_map.csv`) recording how each type was inferred (decimals > 0 → NUMERIC; name/length heuristics otherwise; legacy 8-digit dates flagged). Logical files become indexes/views over their physical file. The mapping register is the review artifact: a data architect signs off field-by-field before production load.

Step 2 — Reference data first

Load the code tables before any transaction data: governorates, regional centres, centres, villages and dependencies (the XLS exports already provide these in full), education stages, dependency/ownership/condition tables, item catalogue (DES*), suppliers (PBYSUPL). The building register follows; everything else keys off it.

Step 3 — ETL from the live machine

- Extract from Db2 for i on the AS/400 (SQL/ODBC or save-file/CSV export per file), landing raw extracts in staging tables that mirror the legacy layout 1:1.
- **Arabic codepage recovery:** Arabic column headings and text data must be extracted with the correct CCSID (the PDF listings render Arabic as mojibake — a print artifact, not data loss). Extract with CCSID 420/425 → UTF-8 conversion and validate against the XLS exports, which decode cleanly via CP1256.
- Transform: packed-decimal and zoned numerics → native types; YYYYMMDD integers → real dates with validity checks (the legacy CHKDAT routine's rules); code values validated against the reference tables; rejected rows go to an exception queue with reasons — never silently dropped.
- Reconciliation: row counts and control totals per file (the legacy system's own summary roll-ups, e.g. SUM408R/SCHSUMPF statistics and the HQ schools-vs-errors statement, are the natural check figures), plus the area-vs-survey ±10% control for buildings.

Step 4 — Historical depth decision

Open items (current buildings, live tenders, unpaid abstracts, active expropriation cases, current fiscal-year budget) migrate in full. Closed historical years can migrate as read-only

archive schemas — cheaper, and keeps the operational database lean. This is a business decision for the Authority, priced both ways.

5. Application rebuild — module by module

Module	Rebuilt from	Key behaviour preserved
Reference & Geography	PBLTAB*, XLS code tables	Full coded geography; 45 stages; ownership/condition/ dependency tables; single maintenance point for all modules.
School Map & Planning	BLCD01AR hub + BLCD02–24 sub-programs; PLN*/PLAN*; SUM408R	Composite building record with annexes/rooms/utilities/ coordinates; check-digit validation; data-quality error report (BLCD30AR) becomes a live dashboard; plan targets, needs and solutions; statistics recomputed, never typed.
Budget	CBA009OR family; CBABDALW/CBABGALW; CBA333*/334* reports	One entry module for payments/authorizations/ assignments; validation against governorate, work-code, funding-source and budget tables before posting; statements generated from posted data (the legacy REPAIR* aggregate-then-print separation, kept).
Purchases & Contracts	#BY001–#BY054 stage programs; PBYSUPL	Tender lifecycle as a workflow with the same gates: booklet, bids, recorded exclusion, envelope committee, financial evaluation, award committee, negotiation, award; deadline checks (CHK5DAYS) automated with alerts.
Estimates (BOQ)	PMPITMAR; PMPT200R4; PMP55555	Catalogue-only item picking (CAT005RR); draft vs. reviewed estimate states with separate approval; priced BOQ PDF with computed totals.
Technical Abstracts	INV001AR1/INV005AR; INM*; PAG*; BIL*	Quantity × price line valuation; typed deductions with distinct calculation per kind; certification → billing → payment as three controlled steps with status workflow.
Real Estate	REMV01/02AR; RNT001AR; LND05/13/26AR; GRN002AR	Rented-building acquisition pipeline as a tracked workflow; expropriation locked until the decree is recorded, then decision/Gazette/display-lists/sale-forms/compensation captured; land parcels with borders and committee approvals; approvals issued as printed documents; aging report for unfinished cases (1–2 / 2–3 / >3 years).

6. Automation the new system adds

- **Scheduled jobs replace batch RPG:** planning refresh (was PLN105AR), statistics roll-ups (SUM408R), aging and data-quality scans run as monitored jobs with logs — the POC's Migration Center job runner shows the pattern.
- **Deadline & workflow automation:** tender deadlines, lease expiries (RNT* monthly position), expropriation stage aging, and plan-target tracking raise tasks/alerts instead of waiting for someone to print a report.

- **Approvals with audit:** committee-driven approvals (land, expropriation, tender award) become tracked approval chains — who approved, when, on what data — replacing signature-chasing across paper and screens.

7. Delivery plan (phased, no big-bang)

Phase	Content	Exit criteria
0 — Foundation	Reference/geography data, building register, users/roles, Migration Center. Schema sign-off on field_map.csv.	Register reconciles with AS/400 counts; Authority data architect signs the mapping register.
1 — School Map	Building composite record, annexes/ utilities/coordinates, data-quality dashboard, planning targets.	Parallel run: new registry vs. BLCD30AR error report and HQ status statement agree.
2 — Money chain	Estimates → Abstracts → Budget, in that order (the legacy file flow), with Purchases supplier master.	One full project lifecycle (estimate → certificate → payment) reconciles end-to-end against legacy totals.
3 — Purchases & Real Estate	Tender workflow; rentals pipeline; expropriation; land.	Live tender and one expropriation case run fully in the new system; aging report matches legacy.
4 — Cutover	Final delta migration, AS/400 switched to read-only archive, hypercare.	Reconciliation pack signed; legacy retained read-only for audit.

Each phase runs in parallel with the AS/400 until its exit criteria pass. Nothing switches off on a date — it switches off on evidence.

8. Gaps and risks — stated plainly

> **Missing from the delivery:** (1) DDS for the shared `PBA3*` Budget master files — only logical files over them were included; their layouts must be obtained from the live system before Phase 2. (2) Display-file DDS behind the workstation files is visible only through RPG usage. (3) ~360 screen screenshots in the DOCX specs were sampled (15 read); a full screen-by-screen transcription needs Arabic OCR — recommended for the requirements baseline, not a blocker for Phases 0–1.

- **Arabic text integrity** is the top data risk: all extraction must be CCSID-validated; the POC's mojibake headings show exactly what goes wrong otherwise. Mitigation is procedural (Step 3) and testable on day one.
- **Business rules live in code, not documents.** They are being extracted rule-by-rule from the RPG (this blueprint's module table is the first pass). Budget the discovery effort per module; the cloned families reduce it — one family's rules cover ~9 legacy programs.
- **Two XLS exports hit the 16,383-row export cap** (village dependencies) — true counts must come from the live files.

- **Data quality:** the legacy system itself reports schools with erroneous data (BLCD30AR) and even the owner reference table contains a stray school name. Migration must include a cleansing queue owned by the Authority's data team, not silent fixes.

9. What the POC already proves

> Delivered with this blueprint: (1) the complete parsed legacy inventory (502 members; every file, field, key and program-file relationship); (2) generated PostgreSQL DDL for 125 tables with the field-mapping register; (3) a working bilingual web application — all six modules with real data storage, roles and audit, plus the Migration Center (legacy inventory browser, field-mapping viewer, ETL job runner that loads staging rows into the modern tables and reports rows read/loaded/rejected). The approach is demonstrated end-to-end on the Authority's real structures, not on invented examples.

Sources: 502 SEU source listings (IBM i V6R1M0, NEWLIBGAEB), 6 DOCX screen inventories, 4 PPTX ERD diagrams, 42 XLS reference-data exports — all from the client's delivered archive, parsed and cross-checked November 2026. Figures quoted are counts from that parse.